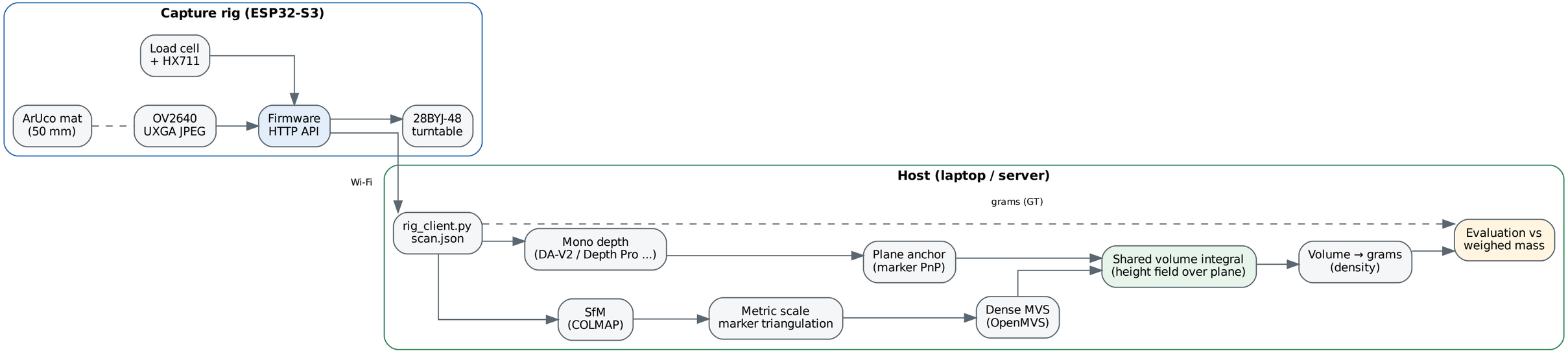
Capture rig (ESP32-S3)
Load cell + HX711
ArUco mat (50 mm)
OV2640 UXGA JPEG
Firmware HTTP API
28BYJ-48 turntable
Wi-Fi
Host (laptop / server)
grams (GT)
rig_client.py scan.json
Mono depth (DA-V2 / Depth Pro ...)
Plane anchor (marker PnP)
Shared volume integral (height field over plane)
Volume → grams (density)
Evaluation vs weighed mass
SfM (COLMAP)
Metric scale marker triangulation
Dense MVS (OpenMVS)